Topic 9: Introducing Evaluation (part 1)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
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OVERVIEW

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Why, What, Where, and When to evaluate

Iterative design and evaluation is a continuous process:

- **Why:** To check that all aspects of the user experience are taken into account in design
- **What:** A conceptual model, early and subsequent prototypes of a new system, more complete prototypes, and to compare a design with competitors' products
- **Where:** In natural, in-the-wild, in living lab and lab settings
- **When:** Throughout design; finished products can be evaluated to collect information to inform new products and for comparisons with other products

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TYPES OF EVALUATION

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Evaluation Types

- **Controlled settings** that directly involve users (for example, usability and research labs)
- **Natural settings** involving users (for instance, online communities and products that are used in public places)
- Often there is little or no control over what users do, especially in in-the-wild settings
- Any **setting that doesn't directly involve users** (for example, consultants and researchers critique the prototypes, and may predict and model how successful they will be when used by users

- Remote evaluation can be done with studies in all three categories above.
- Remote evaluation was practiced since the 1990s but it became important recently for coping with the impact of the Covid-19 pandemic
- Different methods my also be used together to provide different perspectives. For example, testing in a lab maybe used to compare different parts of a design, such as two different menu structures. Then the design may be tested in a natural setting to see whether people like using it in their homes or in other natural settings

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Living Labs

- People's use of technology in their everyday lives can be evaluated in living labs. For example, evaluators can see how people use technology over a period of time, and how it fits in/or not with their routines
- Such evaluations are too difficult to do in a usability lab
- An early example was the Aware Home that was embedded with a complex network of sensors and audio/video recording devices (Abowd et al., 2000)

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Living Labs (cont.)

- More recent examples include whole blocks and cities that house hundreds of people, for example, Verma et al., research in Switzerland (2017), and the huge PEARL facility in the UK
- Many citizen science projects can also be thought of as living labs, for instance, iNaturalist.org
- These examples illustrate how the concept of a lab is changing to include other spaces where people's use of technology can be studied in realistic environments

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EVALUATION
CASE STUDIES

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Constrasting Case Studies

- A study to test how well an automated vehicle supports drivers' behavior as they transition control to and frow between the automated vehicle and the drive.
- An ethnographic study of visitors at the the Royal Highland show in which participants are directed and tracked using a mobile phone app
- Crowdsourcing in which the opinions and reactions of volunteers (for example, from the crowd) inform technology evaluation

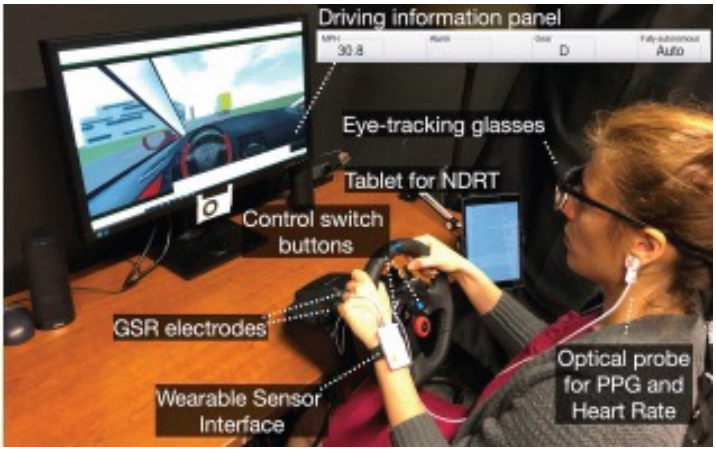
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DeepTake and Automated Vehicles

- Large amounts of data were collected about drivers' driving and non-driving behavior and analyzed using machine learning techniques
- A user study was conducted to examine driver takeover behavior – ie, when a driver took control of the vehicle and when they left control to the automated system
- 11 females and 9 males aged 18-30 years performed tasks of varying difficulty
- Each task took about an hour and was followed up by a questionnaire. Eye tracking was used to track where the driver was looking and the time and processes to complete each task was recorded

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The user study set-up

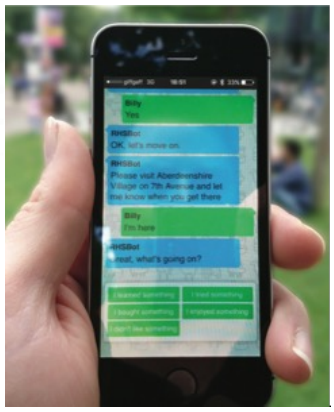


Source: Pakdamanian et al., 2021. CHI'2021 https://dl.acm.org/doi/pdf/10.1145/3411764.3445563

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Ethnobot: Gathering Ethnographic data at the Royal Highland Show

- The Ethnobat directed Billy to a particular place (Aberdeenshire Village)
- Next, Ethnobot asks "…what's going on?"
- The screen shows five of the experience buttons from which Billy needs to select a response

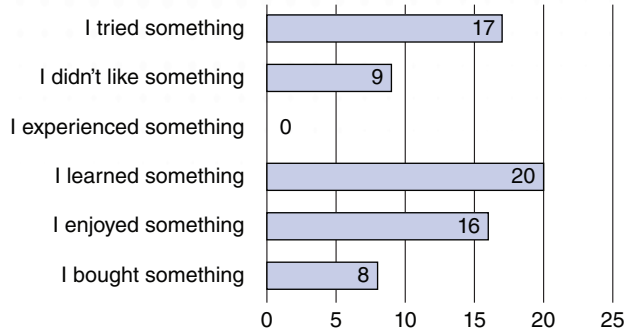


Source: Tallyn et al. (2018) Reproduced with permission of ACM Publications.

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Experience responses to questions submitted in Ethnobot

Number of prewritten experience responses submitted by participants to the pre-established questions that Ethnobot asked them about their experiences



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Mechanical Turk Test - Amazon



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What did we learn from the case studies?

- How to observe users in a lab simulation and in natural settings
- How evaluators use different levels of control in the lab and in natural settings and in crowdsourcing evaluation studies
- Use of different evaluation methods
- How to develop different data collection and analysis techniques to evaluate user experience goals such as engagement
- The ability to run experiments on the Internet that are quick and inexpensive using crowdsourcing
- How a large number of participants can be involved in crowdsourcing using systems such as Mechanical Turk

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Evaluation Methods

Method	Controlled settings	Natural settings	Without users
Observing	X	X	
Asking users	X	X	
Asking experts		X	X
Testing	X		
Modeling			X

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The Language of Evaluation

- Analytics
- Analytical evaluation
- Biases
- Controlled experiment
- Crowdsourcing
- Distributed evaluation
- Ecological validity
- Expert review
- Field study
- Formative evaluation
- Heuristic evaluation
- Informed consent
- In the wild evaluation
- Living lab
- Pain points
- Predictive evaluation
- Reliability
- Research in the wild studies
- Scope
- Summative evaluation
- Usability lab
- User studies
- Usability testing
- Users or participants
- Validity

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OTHER ISSUES TO CONSIDER

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Participants' Rights and Getting Consent

- Participants need to be told why the evaluation is being done, what they will be asked to do and informed about their rights
- Informed consent forms provide this information and act as a contract between participants and researchers
- The design of the informed consent form, the evaluation process, data analysis, and data storage methods are typically approved by a higher authority.

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Things to Consider when Interpreting Data

- **Reliability:** Does the method produce the same results on separate occasions?
- **Validity:** Does the method measure what it is intended to measure?
- Ecological validity: Does the environment of the evaluation distort the results?
- **Biases:** Are there biases that distort the results?
- **Scope:** How generalizable are the results?

Summary

- Evaluation and design are very closely integrated
- Some of the same data gathering methods are used in evaluation as for establishing requirements and identifying users' needs, for example, observation, interviews, and questionnaires
- Evaluations can be done in controlled settings such as lab, less controlled field settings, and remotely or without users being involved
- Usability testing and experiments enable the evaluator to have a high level of control over what gets tested, whereas evaluators typically impose little or no control on participants in field studies, particularly in in the wild studies
- Different methods can be combined to get different perspectives
- Participants need to be made aware of their rights
- It is important not to over-generalize findings from an evaluation